

Student Enrollment & Course Management System (SECMS)

User Story:

- The Student Enrollment & Course Management System is a Salesforce-based application developed to simplify and automate the academic administration processes of an educational institution. The system provides a centralized platform for administrators, enrollment staff, and instructors to manage student information, courses, enrollments, instructor allocations, approval workflows, fee payments, and academic reports efficiently. By bringing these activities together in a single system, the application improves operational efficiency, reduces manual work, and ensures better coordination across different academic functions.

1. INTRODUCTION

1.1 Project Overview :

1. This project is designed to bring student, course, and enrollment information together within a centralized Salesforce application, providing easy access to data, organized processes, and reliable reporting. The system enables Enrollment Officers to efficiently handle student admissions, course enrollment, and related communication, while instructors and administrators can easily monitor course assignments and student academic activities. The application also supports automated instructor allocation, task generation, data validation, and multiple levels of approval to simplify and control the enrollment process.
2. Using Salesforce Flows, Approval Processes, Validation Rules, Reports, and Dashboards, the institution can manage the complete enrollment lifecycle in an organized manner, starting from student admission through course completion. The system also provides continuous visibility into important performance indicators, including total enrollments, revenue generated, student status, course participation, and outstanding approvals.

1.2 Purpose :

- The primary purpose of SECMS is to replace manual and fragmented enrollment processes with a scalable, automated, and secure Salesforce solution that ensures:
- Accurate student data management
- Structured enrollment workflows
- Automated approvals and validations.

2. IDEATION PHASE

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Enrollment Officer	Manage the student enrollme nt	Enroll Manageme nt scattered	System manual in unintergir e	frustrated and error-prone
PS-2	Administrator	Data is centreize d	reporting is delayed	data is not centralized	Unwhile in making decision

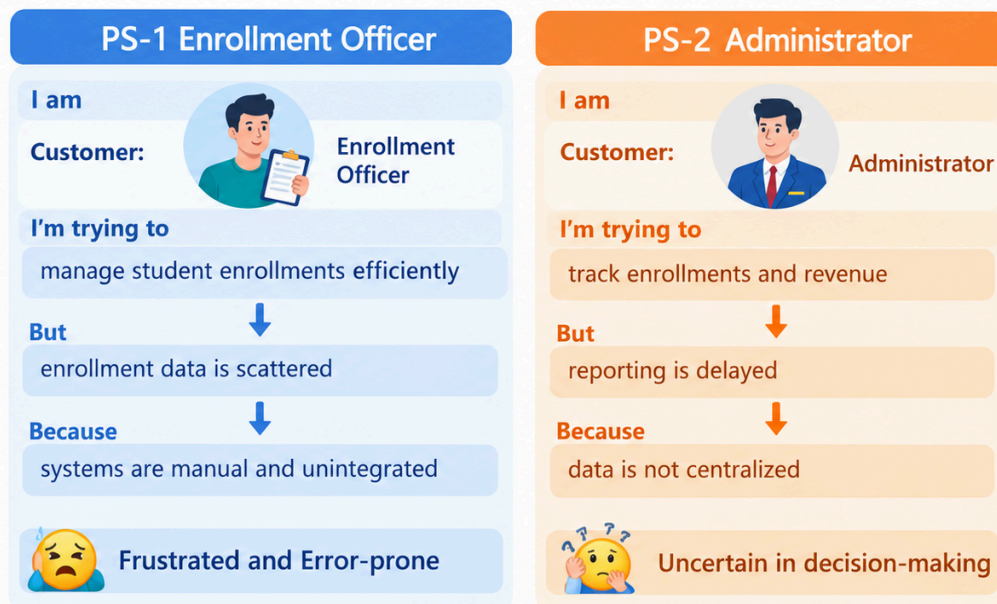
2.1 Problem

Statement (Customer Problem Statement Template)

2. IDEATION PHASE

2.1 Problem Statement

2 Customer Problem Statement Template)



2.2 Empathy Map Canvas

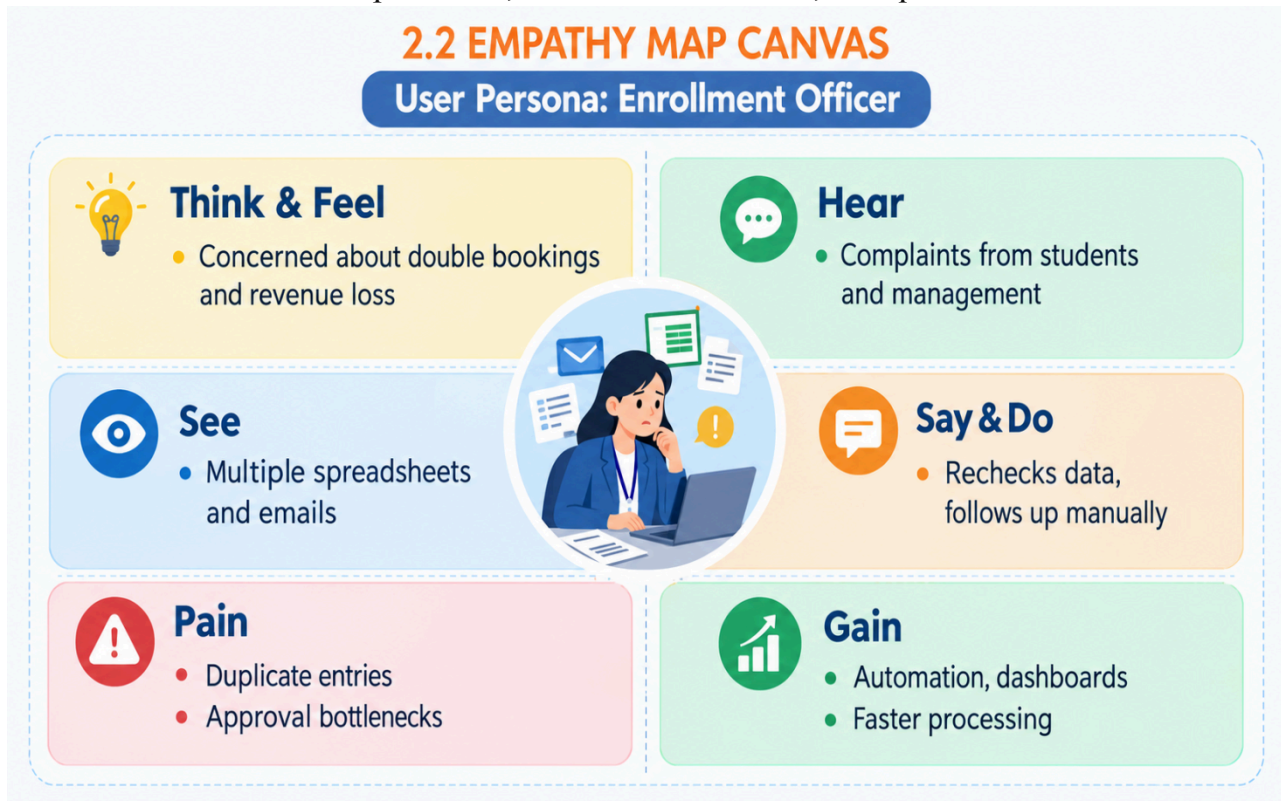
User Persona: Enrollment Officer

Think & Feel: Worried about booking conflicts and financial losses

• **Hear:** Feedback and complaints from students and management

• **See:** Numerous spreadsheets, records, and emails

- **Say & Do:** Verifies information repeatedly and coordinates manually
- **Pain:** Repeated entries, delays in approvals, and data inconsistencies
- **Gain:** Automated processes, real-time dashboards, and quicker enrollment



2.3 Brainstorming & Idea Prioritization

Sure. Here is the revised content with the same format:

1. Step 1 – Brainstormed Ideas

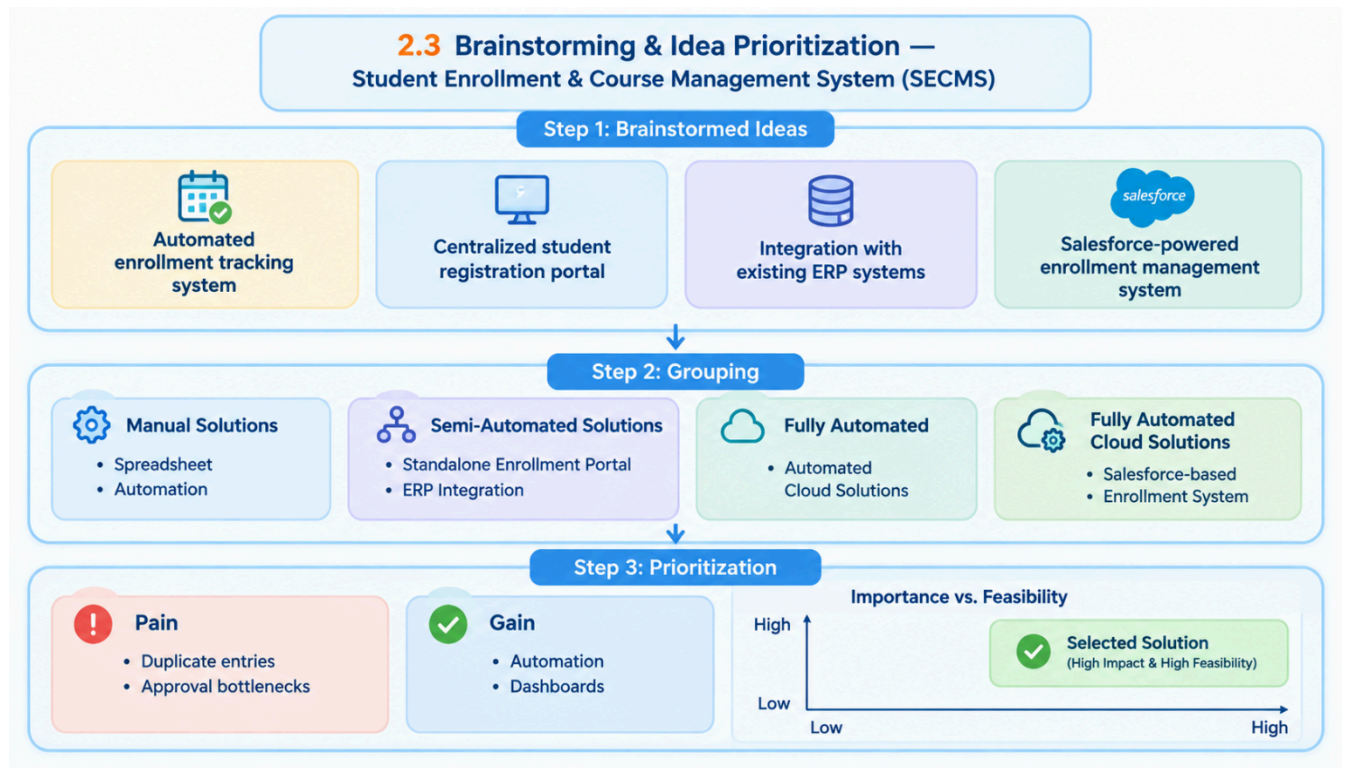
- Automated enrollment tracking system
- Centralized student registration portal
- Integration with existing ERP systems
- Salesforce-powered enrollment management system

Step 2 – Grouping

- Manual solutions
- Semi-automated solutions
- Fully automated cloud solutions

Step 3 – Prioritization

Salesforce-based SECMS was selected due to **high feasibility** and **high impact**.



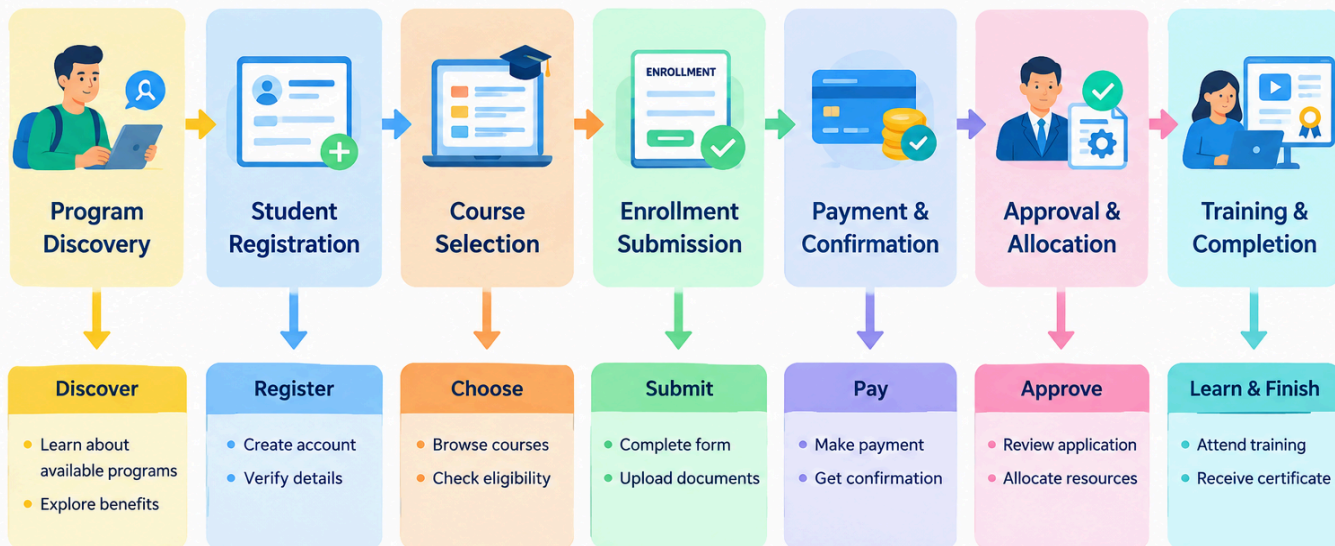
3.REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Stages covered:

- Program Discovery
- Student Registration
- Course Selection
- Enrollment Submission
- Payment & Confirmation
- Approval & Allocation
- Training & Completion

3.1 Customer Journey Map – Student Enrollment & Course Management (SECMS)



3.2 Solution Requirements

Functional Requirements

FR No	Epic	Description
FR-1	Student Management	Manage student records
FR-2	Course Management	Create and manage courses
FR-3	Enrollment Processing	Request, approve, reject enrollments

FR-4	Instructor Assignment	Auto-assign instructors
FR-5	Reporting	Dashboards and reports

Non-Functional Requirements

NFR-1: Usability

Revised Content

- The system shall provide a clear, responsive, and easy-to-use interface that allows enrollment officers,

administrators, instructors, and students to complete their tasks quickly and accurately. The interface shall use structured menus, intuitive dashboards, and simplified forms to minimize complexity, reduce errors, and improve the overall user experience.

- **Examples:**
 - Simplified student profile creation
 - Fast enrollment request processing
 - Personalized dashboards based on user roles

NFR-2: Security

Revised Content

- The system shall maintain secure and controlled access to student and institutional information by implementing authentication, role-based permissions, and data protection mechanisms. Confidential information such as student profiles, course details, payment records, and enrollment information shall be accessible only to users with the appropriate permissions.
- **Examples:**
 - Administrators can manage user accounts and system settings
 - Instructors can access information related to their assigned courses
 - Students can view and update only their own profiles

NFR-3: Reliability

Revised Content

- The system shall provide reliable and stable performance during student registration, enrollment, payment processing, and report generation. All system operations shall be processed accurately and consistently to minimize errors, data loss, and interruptions.
- **Examples:**
 - Prevents repeated enrollment submissions
 - Automatically reflects the latest enrollment status
 - Generates accurate and consistent reports

NFR-4: Performance

The system shall provide fast response times for all operations to ensure smooth and uninterrupted user experience, especially during peak admission periods.

Examples:

- Quick dashboard refresh
- Fast search for student records
- No lag during bulk uploads

NFR-5: Scalability

Revised Content

- **The system shall be designed to accommodate increasing numbers of students, courses, instructors, departments, and administrative users while maintaining efficient performance. The architecture shall also support the integration of additional functionalities and services as organizational requirements evolve.**
- **Examples:**
 - **Support additional courses and programs**
 - **Accommodate growing student registrations**
 - **Integrate new features and external platforms**

NFR-6: Availability

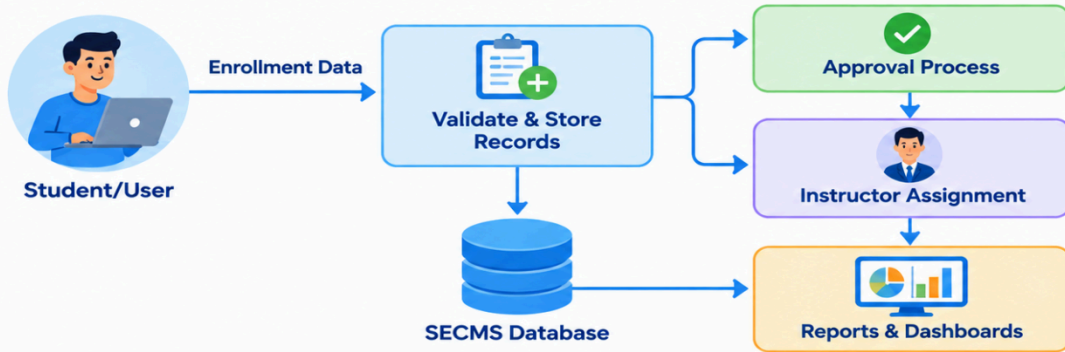
Revised Content

- The system shall provide dependable and uninterrupted access to authorized users during regular institutional operations. It shall maintain service availability during high-demand periods and include mechanisms to protect data from unexpected system failures.
- **Examples:**
 - Continuous access to enrollment services
 - Stable performance during peak registration periods
 - Scheduled backup and recovery of important records

3.3 Data Flow Diagram

1. ● Users enter student, course, and registration details
 - The system verifies and saves the submitted information
 - Enrollment requests are forwarded to the appropriate authorities
 - Approved data is used for instructor assignment and reporting
 - Dashboards display updated enrollment and course information

3.3 Data Flow Diagram – Student Enrollment & Course Management System (SECMS)



Layer	Technology
UI	Salesforce Lightning
UI Salesforce Lightning	
Logic	Salesforce Flows
Database	Salesforce Objects
Automation	Validation Rules, Approval Process

3.4 Technology Stack

Layer	Technology
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UI Salesforce Lightning

Logic	Salesforce Flows
Database	Salesforce Objects
Automation	Validation Rules, Approval Process
Reporting	Reports & Dashboards

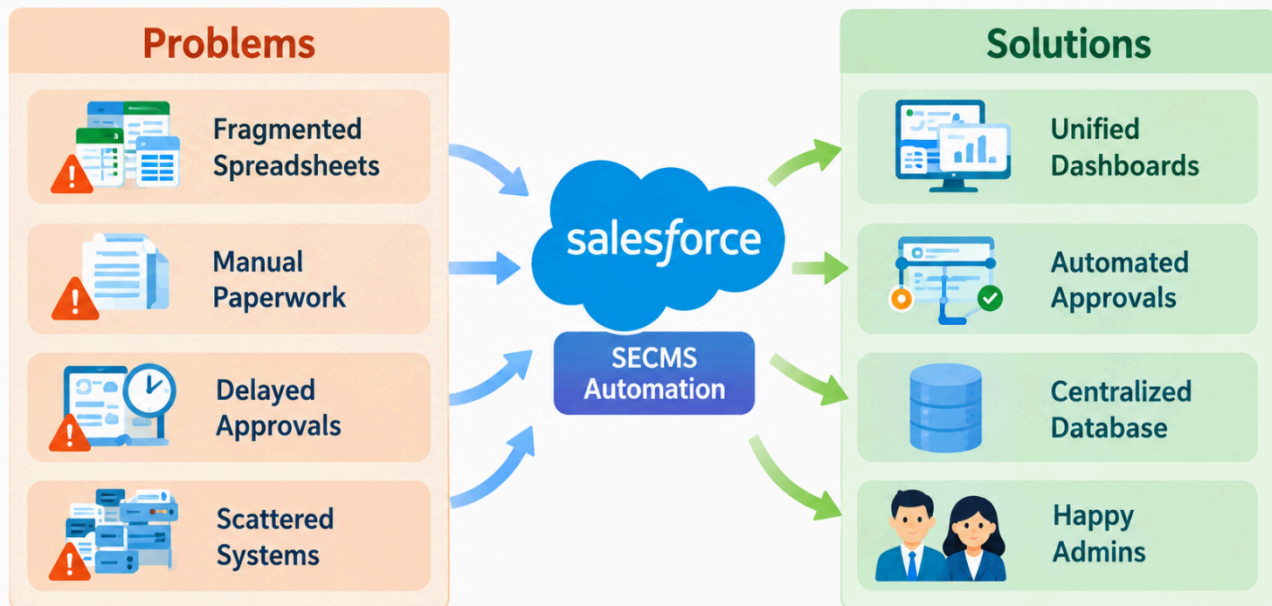
4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

SECMS directly addresses fragmented enrollment workflows by aligning user behavior, constraints, and automation using Salesforce-native capabilities.

4.1 Problem-Solution Fit

— Student Enrollment & Course Management System (SECMS)



4.2 Proposed Solution

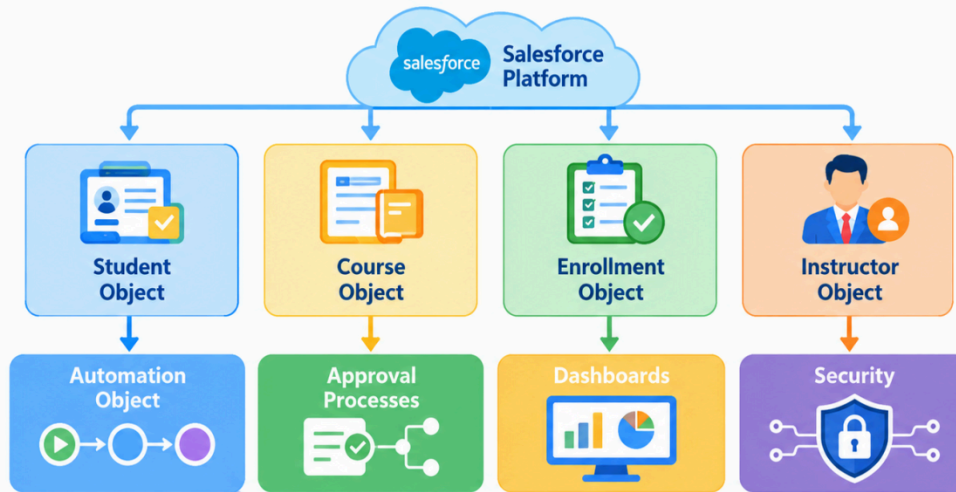
Functional Requirements

Parameter	Description
Problem	Manual enrollment handling
Solution	Salesforce-based SECMS
Novelty	End-to-end automation
Social Impact	Improved student experience
Revenue Model	Institutional deployment
Scalability	Multi-department support

4.3 Solution Architecture

The architecture integrates Salesforce objects, automation flows, approval processes, dashboards, and security layers.

4.3 Solution Architecture



5.1 Project Planning

- Agile methodology
- Epics → Stories → Story Points
- Sprint-based execution
- Velocity calculation used for estimation

5.

PROJECT PLANNING & SCHEDULING

5.1 Project Planning

- Agile methodology
- Epics → Stories → Story Points
- Sprint-based execution
- Velocity calculation used for estimation

1. Product Backlog, Sprint Schedule, and Estimation

2.

3.

Sprint-1	Salesforce Environment setup	As a administrator,I want to create a objects	1	High	Member 1
Sprint-1	Data architecture	As administrator create a objects for student	3	High	Member2
Sprint-2	Navigation setup	As a user ,Iwant separate navigation setup of student	2	Medium	Member3

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Date)	Planned End Date	Sprint Release Date (Actual)
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Sprint-1	20	6 Days	24 Oct 2022	29 Oct 2022	20	29 Oct 2022	Sprint-2	20	6 Days	31 Oct 2022	05 Nov 2022
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Sprint-3	20	6 Days	07 Nov 2022	12 Nov 2022		
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Sprint-4	20	6 Days	14 Nov 2022	19 Nov 2022		
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6. Project Development Phase:

Project Flow:

Revised Milestones

- **Milestone 1:** Salesforce Environment Setup
- Milestone 2:** Student and Course Data Structure Setup
- Milestone 3:** Navigation Tab Configuration
- Milestone 4:** Student Enrollment Lightning Application Setup
- Milestone 5:** Student, Course, Enrollment, and Instructor Field Configuration
- Milestone 6:** Object Relationships and Data Linking
- Milestone 7:** Record Page and Layout Customization
- Milestone 8:** Data Validation and Quality Controls
- Milestone 9:** Enrollment Workflow Automation
- Milestone 10:** Enrollment Review and Approval Workflow
- Milestone 11:** Enrollment Reports and Data Analysis
- Milestone 12:** Performance Dashboards and Visual Analytics
- Milestone 13:** User Access and Security Configuration

Milestone 14: Conclusion

What you'll learn

1. Real-time Salesforce project implementation
2. Designing custom objects and relationships
3. Creating page layouts & record types
4. Writing validation rules
5. Building automation with flows
6. Creating approval workflows
7. Developing reports & dashboards
8. Setting up security with profiles, permission sets, and sharing
9. End-to-end application testing

Milestone 1-Salesforce :

Introduction:

Are you new to Salesforce? Not sure exactly what it is, or how to use it? Don't know where you should start on your learning journey? If you've answered yes to any of these questions, then you're in the right place. This module is for you.

Welcome to Salesforce! Salesforce is game-changing technology, with a host of productivity-boosting features, that will help you sell smarter and faster. As you work toward your badge for this module, we'll take you through these features and answer the question, "What is Salesforce, anyway?".

What Is Salesforce?

Salesforce is your customer success platform, designed to help you sell, service, market, analyze, and connect with your customers. Salesforce has everything you need to run your business from anywhere. Using standard products and features, you can manage relationships with prospects and customers, collaborate and engage with employees and partners, and store your data securely in the cloud. So what does that really mean? Well, before Salesforce, your contacts, emails, follow-up tasks, and prospective deals might have been organized something like this:

<https://youtu.be/r9EX3IGde5k>

Activity 1: Creating Developer Account:

Creating a developer org in salesforce.

Go to <https://developer.salesforce.com/signup>

On the sign up form, enter the following details :

10:56 m    

  salesforce.com/products   

Sign up for your Developer Edition

A free Salesforce Platform environment with Agentforce and Data 360

Job title

Student ✓

Country/region

India ✓ ▼

State/province

Tamil Nadu ✓ ▼

Work email

shyam20324u09083@g... ✓

☒ I agree to the [Main Services Agreement – Developer Services](#), [Salesforce Program Agreement](#), and [API Terms of Service](#) 

Your free trial may be provisioned on or migrated to Hyperforce, Salesforce's public cloud infrastructure.

& Last name

Email

Role : Developer

Company : College Name

County : India

Postal Code : pin code

Username : should be a combination of your name and company This need not be an actual email id, you can give anything in the format :

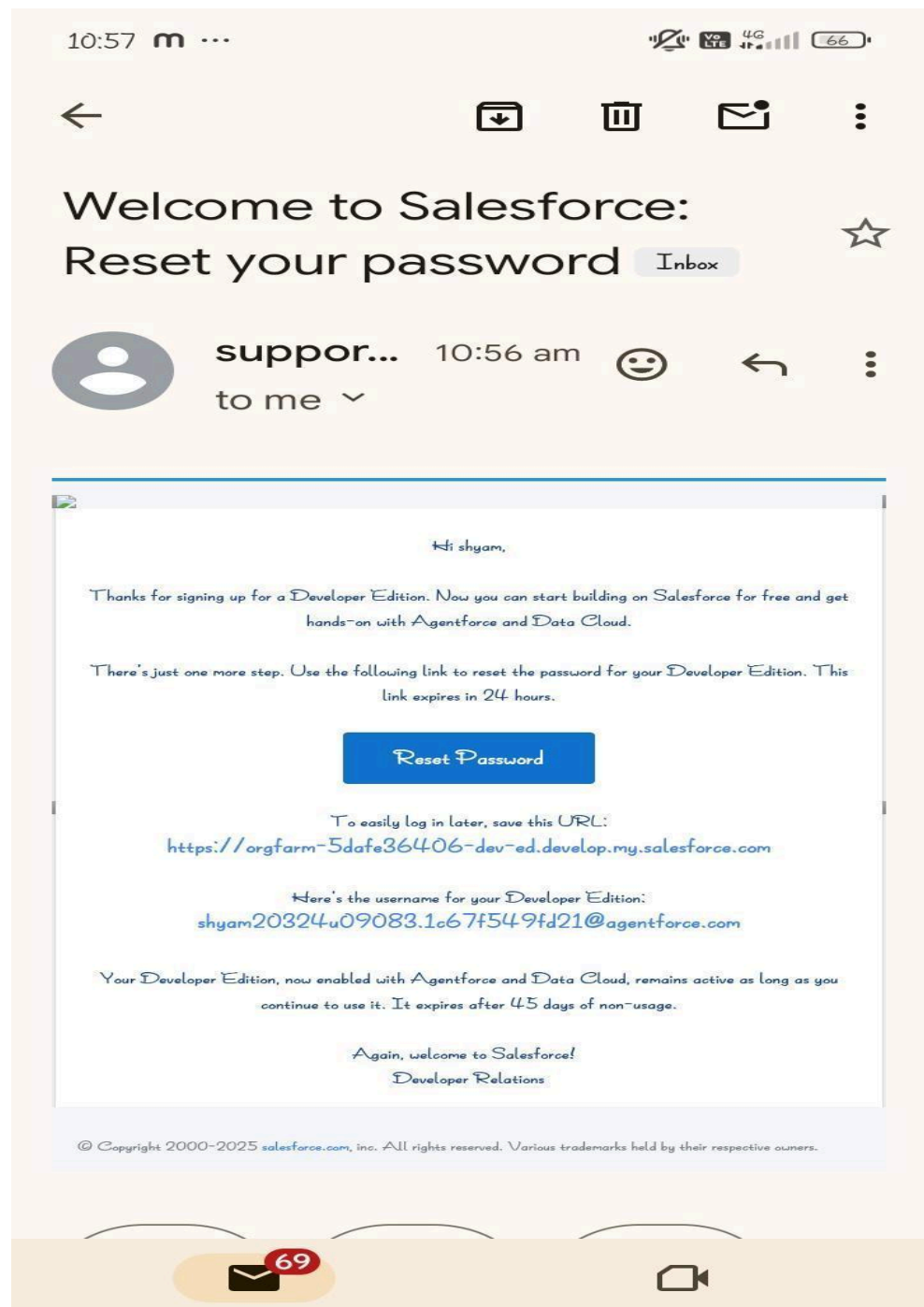
[username@organization.com](#)

First name

Click on sign me up after filling these.

Activity 2: Account Activation:

1. Go to the inbox of the email that you used while signing up. Click on the verify account to activate your account. The email may take 5-10mins.



2. Click on Verify Account

3. Give a password and answer a security question and click on change password.

11:00 m ... 4G LTE 65%

Change Your Password

Enter a new password for
shyam20324u09083.1c67f549fd21@agentforce
Make sure to include at least:

- ✓ 8 characters
- ✓ 1 letter
- ✓ 1 number

* New Password

..... Good

* Confirm New Password

..... Match

* New Security Question

▼ In what city were you born?

* New Answer

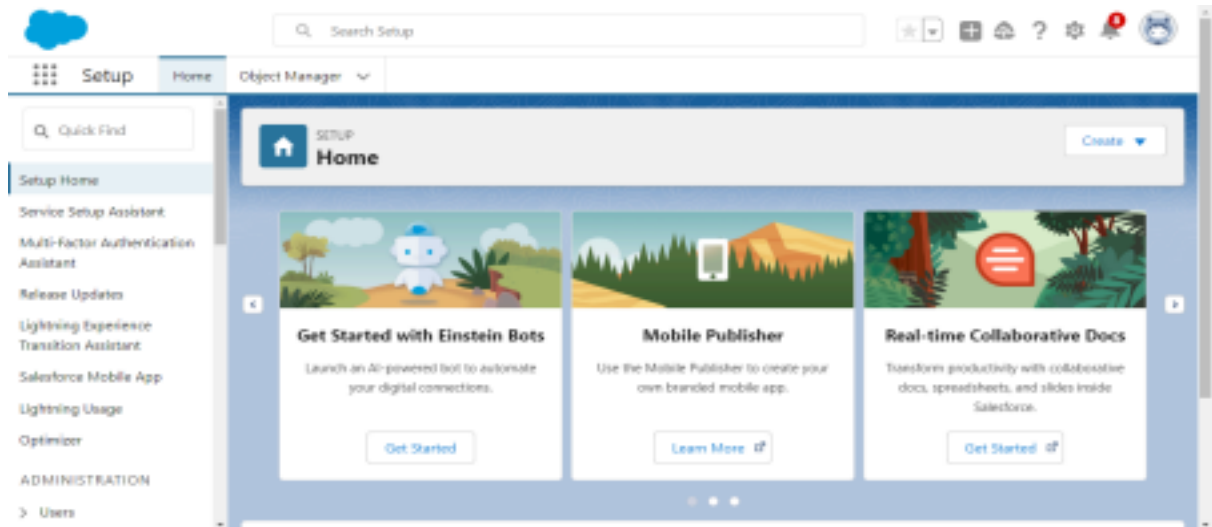
asdfghjkl Good

*=required

Change Password

Password was last changed on

4. Then you will redirect to your salesforce setup page.



Milestone 2 – Objects:

Activity 1: Creating Student Object:

The purpose of creating the Student custom object is to store and manage information about students such as their personal details, contact information, and enrollment status.

To create the Student object:

Go to the Setup page.

Click on Object Manager.

Click on Create → Custom Object.

Enter the details as below:

1. Label: Student
2. Plural Label: Students
3. Object Name: Student
4. Record Name: Student Name
5. Data Type: Text
6. Under Optional Features, select:
7. Allow Reports
8. Allow Activities (optional but recommended)
9. Track Field History (optional, if you want)
10. Under Search Status, select:
11. Allow Search
12. Click Save.

Setup Home Object Manager

SETUP > OBJECT MANAGER

Student

Details

- Fields & Relationships
- Page Layouts
- Lightning Record Pages
- Buttons, Links, and Actions
- Compact Layouts
- Field Sets
- Object Limits
- Record Types
- Related Lookup Filters
- Restriction Rules

Custom Object Definition Edit Save Save & New Cancel

Custom Object Information

The singular and plural labels are used in tabs, page layouts, reports, and alerts. Be careful when changing the name or label as it may affect existing integrations and merge templates.

Label Example: Account

Plural Label Example: Account

The Object Name is used when referencing the object via the API.

Object Name Example: Account

Description

Context-Sensitive Help Setting ☒ Open the standard Salesforce.com Help & Training window ☐ Open a window using a Visualforce page

Content Name

Setup Home Object Manager

SETUP > OBJECT MANAGER

Student

Details

- Fields & Relationships
- Page Layouts
- Lightning Record Pages
- Buttons, Links, and Actions
- Compact Layouts
- Field Sets
- Object Limits
- Record Types
- Related Lookup Filters
- Restriction Rules
- Scoping Rules

Enter Record Name Label and Format

The Record Name appears in page layouts, key lists, related lists, lookups, and search results. For example, the Record Name for Account is "Account Name" and for Case is Case Number". When selecting a name format, use the drop-downs to specify the label and format.

Record Name Example: Account Name

Data Type Example: Account

Optional Features

☒ Allow Reports

☒ Allow Activities

☐ Track Field History

☐ Allow in Chatter Groups

☐ Enable Licensing

Object Classification

When these settings are enabled, this object is classified as an Enterprise Application object. When these settings are disabled, this object is classified as a Light Application

☒ Allow Sharing

☒ Allow Bulk API Access

☒ Allow Streaming API Access

Deployment Status

☒ In Development ☐ Deployed

Deployment Status

☐ In Development ☒ Deployed

Search Status

When this setting is enabled, your users can find records of this object type when they search. [Learn more.](#)

☒ Allow Search

Save Save & New Cancel

In the same way Create **Courses, Instructors, and Enrolments** objects

Milestone 3 - Fields:

Table 1: Student Object Fields:

Field Label	Data Type
-------------	-----------

Student Name	Text (Standard)		
Email	Email		
Phone	Phone		
Date of Birth	Date		
Student Status	Picklist		
Address	Text Area		
Field Label	Data Type		

Table 3: Instructor Object Fields:

Field Label	Data Type
Instructor Name	Text (Standard)
Instructor Code	Text
Expertise	Picklist
Phone	Phone
Email	Email

Table 4: Enrollment Object Fields:

Field Label	Data Type
Student	Lookup (Student)
Course	Lookup (Course)

Course Name	Text (Standard)	
Category	Picklist	
Duration	Number/Text	
Course Fee	Currency	
Description		

Table 2: Course Object Fields:

Instructor	Lookup (Instructor)
------------	------------------------

Enrollmen
t Date
Date

Enrollmen
t Status
Picklist

Fees Paid	Checkbox
Total Amount	Currency
Comme nt s	Text Area

Milestone 4 – Create Relationships:

1. Activity 1: Create Lookup from Enrollment to Student:

1. Go to Setup → click Object Manager.
2. Type Enrollment in Quick Find → click Enrollment__c object.
3. Click Fields & Relationships → click New.
4. Select Lookup Relationship → click Next.
5. For Related To, select Student__c → click Next.
6. Enter Field Label: Student (API Name: Student__c).
7. Click Next through Field-Level Security and Layout options (ensure it's visible and added to layout).
8. Click Save.

SETUP > OBJECT MANAGER Help for this Page ?

Student

Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Restriction Rules
Scoping Rules

Student

The Student object represents student records in the system. It contains information such as student name, contact details, course enrollment and academic history.

Details			
API Name	Student__c	Description	Stores information about students enrolled in courses.
Label	Student	Object Classification	Custom Object
Plural Label	Students		

Standard Fields & Relationships

The standard fields and relationships that can be used with this object are shown below.

No standard fields at this time.

Field Label: ⓘ
Field Name: ⓘ
Description:
Help Text: ⓘ
Child Relationship Name: ⓘ
- Do if the lookup record is deleted?
☐ The relationship is required on the child object, and it's required if you make this field required.
☐ Clear the value of this field. You can't choose this option if you make this field required.
☒ Don't allow deletion of the lookup record that's part of a lookup relationship.

Previous Save Next Cancel

Activity 2: Create Lookup from Enrollment to Course

Purpose:

To link each Enrollment to a Course.

Steps:

1. Go to Setup → Object Manager → Enrollment__c.
2. Click Fields & Relationships → New.
3. Select Lookup Relationship → Next.
4. Related To: select Course__c → click Next.
5. Field Label: Course (API Name: Course__c).
6. Click Next (set visibility and layout) → Save.

Edit Enrollment: Custom Field Help for this Page ?

Course

Step 1. Choose the field type Step 1

Next Cancel

Specify the type of information that the custom field will contain.

Data Type
☐ None Selected
☒ Lookup Relationship

Select one of the data types below:
Creates a relationship that links this object to another object. The relationship field allows users to click on a lookup icon to select a value from a list. The other object is the source of the values in the list.

Edit Relationship
Enrollment [Help for this Page](#)

Step 2. Enter the label and name for the lookup field Step 2 of 4

Field Label [i](#)

Field Name [i](#)

Description

Help Text [i](#)

Child Relationship Name [i](#)

What to do if the lookup record is deleted?

- ☒ Do not allow deletion of the lookup record that is part of a lookup relationship.
- ☐ Clear the value of this field. You can't choose this option if you make this field required.

[Previous](#) [Save](#) [Next](#) [Cancel](#)

Activity 3: Create Lookup from Enrollment to Instructor:

Purpose:

To store which Instructor is assigned to a particular Enrollment.

Steps:

1. Go to Setup → Object Manager → Enrollment__c.
2. Click Fields & Relationships → New.
3. Choose Lookup Relationship → Next.
4. Related To: select Instructor__c → Next.
5. Field Label: Instructor (API Name: Instructor__c).
6. Complete Field-Level Security and Page Layout options → Save

SETUP > OBJECT MANAGER

Enrollment

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Specify the type of information that the custom field will contain.

Data Type

☐ None Selected

☐ Auto Number

☐ Formula

☐ Roll-Up Summary [i](#)

☒ **Lookup Relationship**

Select one of the data types below:

A system-generated sequence number that uses a display format you define. The number is automatically incremented for each new record.

A read-only field that derives its value from a formula expression you define. The formula field is updated when any of the source fields are updated.

A read-only field that displays the sum, minimum, or maximum value of a field in a related list or the record count of all records related to the current record.

Creates a relationship that links this object to another object. The relationship field allows users to click on a lookup icon to select a value from the related object. The related object is the source of the values in the list.

Enrollment

New Relationship [Help](#)

Step 2. Choose the related object

Select the other object to which this object is related.

Related To

[Previous](#) [Next](#)

Step 3. Enter the label and name for the lookup field

Field Label: ⓘ

Field Name: ⓘ

Description:

Help Text: ⓘ

Child Relationship Name: ⓘ

What to do if the lookup record is deleted?

- ☒ Do not allow deletion of the lookup record that's part of a lookup relationship.
- ☐ Clear the value of this field. You can't choose this option if you make this field required.

Enrollment

New Relationship

Step 6. Add custom related lists

Field Label: Instructor

Data Type: Lookup

Field Name: Instructor

Description:

Specify the title that the related list will have in all of the layouts associated with the parent.

Related List Label:

Select the page layouts that should include this field. The field will be added as the last field in the first 2-column section of these page layouts. The page layouts that are selected will be added to the list of page layouts that are associated with the field. To change the location of this field on the page, you will need to customize the page layout. To change the values that appear, you will need to customize the Record Types.

Add Related List	Page Layout Name
☒	Instructor Layout

☒ Append related list to users' existing personal customizations

Milestone 5 – Validation Rules:

Activity 1: Student Email Domain Validation

Purpose:

To ensure students use only allowed institutional email addresses.

Steps:

Go to Setup → Object Manager → Student__c.

Click Validation Rules → New.

Rule Name: Student_Email_Domain_Validation.

Error Condition Formula: (example for @student.edu)

- NOT(CONTAINS(Email__c, "@student.edu"))

Error Message: Email must be a valid student institutional email.

Error Location: Field → Email__c.

Click Save.

The screenshot shows the Salesforce Setup interface. The left sidebar contains a search bar and a list of setup categories: Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, Restriction Rules, Scoping Rules, Object Access, Triggers, Flow Triggers, and Validation Rules (which is highlighted). The main content area is titled 'Student' and 'Validation Rule Edit'. It includes fields for 'Rule Name' (Email_Must_Be_Student_Domain), 'Active' (checked), and 'Description'. Below these is the 'Error Condition Formula' section, which includes an example and a tip. The formula entered is 'NOT(CONTAINS(Email__c, \"@student.edu.in\"))'. To the right of the formula is a 'Functions' list with a dropdown menu and a list of functions: ABS, ACOS, ADDMONTHS, AND, ASCII, and ASIN. A 'Check Syntax' button is located below the formula field. At the top right of the 'Validation Rule Edit' section are buttons for 'Save', 'Save & New', and 'Cancel'.

Activity 2: Fees Must Be Paid Before Approval

Purpose:

Prevent Enrollment from being approved until fees are paid.

Steps:

1. Go to Setup → Object Manager → Enrollment__c.
2. Click Validation Rules → New.
3. Rule Name: Fees_Must_Be_Paid_Before_Approval.
4. Error Condition Formula:
AND(
 ISPICKVAL(Enrollment_Status__c, "Approved"),
 Fees_Paid__c = FALSE

)

5.Error Message: Fees must be paid before approving an enrollment.

6.Error Location: Top of Page (or field as needed).

Click Save.

• Milestone 6 – Approval Process

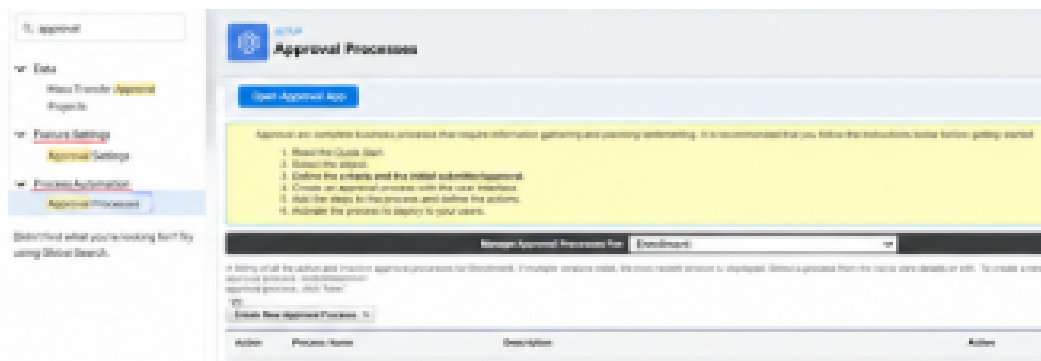
Activity 1: Create Approval Process for Enrollment Requests

Purpose:

Route Enrollment requests to Training Manager for approval or rejection.

Steps:

1. Go to Setup → Approval Processes → select Enrollment__c.
2. Click Create New Approval Process → Use Standard Setup.
3. Process Name: Approval_Enrollment_Request.
4. Entry Criteria: Enrollment_Status__c = "Requested".
5. Save and configure steps.



Approval Enrollment Request

Step 1: Enter Name and Description

Enter a name and description for your new approval process.

Enter Name and Description

Process Name	Approval Enrollment Request
Unique Name	Approval_Enrollment_Request
Description	Approval process for Enrollment requests -- sends to Training Manager

Step 2: Specify Entry Criteria

[Previous](#) [Next](#)

Only certain types of records should enter this approval process, enter the criteria below. For example, only expense reports from employees at headquarters should use this approval process.

Specify Entry Criteria

Use the approval process title following [▼](#)

Field	Operator	Value	
Enrollment Enrollment Status	equals	Requested	AND
--None--	--None--		AND
--None--	--None--		AND
--None--	--None--		AND



SETUP

Approval Processes

Approval Step 1 of 1

Training Manager

[Help for this Page](#)

Step 1: Enter Name and Description

Step 1 of 1

Enter a name, description, and step number for your new approval step.

[Save](#) [Cancel](#)

Enter Name and Description

Name	Training Manager
Unique Name	Training_Manager
Description	

6. Add Approval Step:

7. Approver: specific Training Manager user.

8. Under Final Approval Actions:

9. Add Field Update to set Enrollment_Status__c = "Approved".

Field Updates

Below the field update, including the object associated with the workflow rule, approval process, or entitlement process, the field to update, and the value to apply. Note that the fields are shown only for the type that you select.

Field Update Edit Save Save & New Cancel

Identification

Name:

Unique Name:

Description:

Object: Enrollment

Field to Update: Enrollment: Enrollment Status

Field Data Type: Picklist

Re-evaluate Workflow Rules after Field Change: ☐

Specify New Field Value

Picklist Options:

☐ The value above the current one

☐ The value below the current one

☒ A specific value:

Save Save & New Cancel

10. Under Final Rejection Actions:

11. Add Field Update to set Enrollment_Status__c = "Rejected".

Field Update Edit Save Save & New Cancel

Identification

Name:

Unique Name:

Description:

Object: Enrollment

Field to Update: Enrollment: Enrollment Status

Field Data Type: Picklist

Re-evaluate Workflow Rules after Field Change: ☐

Specify New Field Value

Picklist Options:

☐ The value above the current one

☐ The value below the current one

☒ A specific value:

12. Add Email Alert to send rejection email to student.

The screenshot shows the 'Email Alert Edit' form. At the top, there are three buttons: 'Save', 'Save & New', and 'Cancel'. The form is divided into two main sections: 'Identification' and 'Specify New Field Value'.

Identification Section:

- Name:** Enrollment Rejected Alert
- Unique Name:** Enrollment_Rejected_Alert (with a link icon)
- Description:** (empty text box)
- Object:** Enrollment
- Field to Update:** Enrollment: Enrollment Status
- Field Data Type:** Picklist
- Re-evaluate Worktime Governor Field:** ☐ (with a link icon)
- Change:** (empty text box)

Specify New Field Value Section:

Picklist Options:

- ☐ The value above the current one
- ☐ The value below the current one
- ☒ A specific value: Rejected (with a dropdown arrow)

At the bottom of the form, there are three buttons: 'Save', 'Save & New', and 'Cancel'.

13. Activate the approval process.

Milestone 7- Tabs

Activity 1: Creating a Tab for the Student Object:

Go to Setup.

1. In the Quick Find box, type Tabs.
2. Under Custom Object Tabs, click New.
3. Select Student from the object list.
4. Choose a tab style (icon).
5. Click Next, assign it to all profiles, and then click Save.

The screenshot shows the Salesforce Setup interface. On the left, a navigation menu is visible with the following items: Setup, Feature Settings, Sales, Custom Tabs, External Data Sources, Custom Tabs, User Interface, Custom Code, Custom Tabs, Lightning Components, and Custom Tabs. The main content area is titled 'Publishing Tab' and contains the following fields:

- Label:** Students
- Tab Style:** Group
- Home Page Custom Link:** Home
- Short Description:** Student module for enrollment and course management.

At the bottom of the form, there are 'Save' and 'Cancel' buttons.

- **Activity 2: Creating Remaining Tabs**

Now create the Tabs for the remaining Objects, they are “Courses, Instructors, Enrollment”.

Follow the same steps as mentioned in Activity -1

Milestone 8 - The Lightning App

Activity 1: Creating the Lightning Application for the Project

Navigate to Setup.

Enter App Manager in the Quick Find search box.

Click New Lightning App.

Provide the app details:

App Name: Student Enrollment & Course Management

Developer Name auto-fills

Description: Optional, but can describe the purpose of the app

Click Next and choose the following settings:

App Branding (optional): Add logo or select color theme

Navigation Style: Choose Standard Navigation

Click Next and assign the app to specific profiles (e.g., **Admin, Enrollment Officer, Instructor**).

Click Next to select the items (tabs) to be included in the app. Add:

1. Student
2. Course

3. Instructor
4. Enrollment
5. Reports
6. Dashboards
7. Home
8. Tasks

Click Next and review the app settings.

Finally, click Save & Finish.

The screenshot shows the 'Lightning Experience App Manager' setup interface. The left sidebar contains navigation links: 'Setup', 'Home', and 'Object Manager'. Under 'Setup', there are links for 'Feature Settings', 'Sales', 'Custom Tabs', 'Table', 'External Data Sources', and 'Custom Tasks'. The main content area is titled 'App Details & Branding' and includes a sub-header 'Choose your Lightning app, and manage its description, label, and images. Highlight a color for the navigation bar.' The 'App Details' section has fields for 'Label' (Student Enrollment & Course Management), 'Unique Name' (Student_Enrollment_Course_Management), and 'Description' (Enter a description...). The 'App Branding' section includes an 'Image' field with a placeholder image, a 'Primary Color Hex Value' field with the value #0070C2, and a 'Org Theme Display' checkbox.

Setup Lightning Experience App Manager

App Details & Branding

Choose your Lightning app, and manage its description, label, and images. Highlight a color for the navigation bar.

App Details

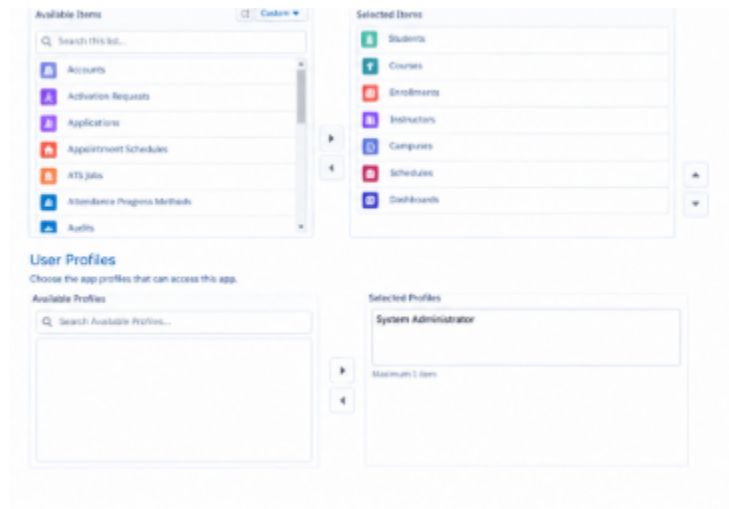
- Label: Student Enrollment & Course Management
- Unique Name: Student_Enrollment_Course_Management
- Description: Enter a description...

App Branding

Image: [Placeholder Image]

Primary Color Hex Value: #0070C2

Org Theme Display: ☐ Use the app's image and color instead of the org's custom theme.



Milestone 9 – Page Layouts and Record:

Activity 1: Create Record Types for Enrollment:

Purpose:

To separate New Enrollment vs Re-Enrollment processes.

Steps:

1. Go to Setup → Object Manager → Enrollment__c.
2. Click Record Types → New.
3. For the first record type:
 - Existing Record Type: Master
 - Record Type Label: New Enrollment
 - Enable for required profiles (Admin, Enrollment Officer).
 - Select the default Page Layout → Save & New.
4. For the second record type:
 - Record Type Label: Re-Enrollment

- Enable for same profiles.
- Select the default Page Layout → Save.

SETUP > OBJECT MANAGER
Enrollment

Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits

Record Types
Related Lookup Filters

Add Record Type
Re-Enrollment
Enter a name for the record type and click Save.

Record Type
Record Type Label: Re-Enrollment
Namespace Prefix: Not applicable
Description: Used when a student re-enrolls in a course.
Active: ☒

Save Cancel

SETUP > OBJECT MANAGER
Enrollment

Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets

Record Types
Related Lookup Filters

Edit Record Type
New Enrollment
Enter a new name for the selected record type and click Save.

Record Type
Record Type Label: New Enrollment
Record Type Name: New_Enrollment
Description: Used when a student re-enrolls in a course.
Active: ☒

Save Cancel

Activity 2: Separate Page Layouts for Each Enrollment Record Type

Purpose:

To show different fields for New Enrollment vs Re-Enrollment.

Steps:

1. Enrollment__c object → click Page Layouts.
2. Click New to create a layout for New Enrollment:
 - Clone from existing Enrollment Layout → name it New Enrollment Layout → Save.
 - Keep fields relevant for first-time enrollment: Student, Course, Instructor, Enrollment Date, Status, Fees, etc.
 - Remove unnecessary re-enrollment fields if any.
 - Click Save.

Setup > OBJECT MANAGER

Enrollment

- Details
- Fields & Relationships
- Page Layouts**
- Lightning Record Pages
- Buttons, Links, and Actions
- Compact Layouts
- Field Sets

Create New Page Layout

As an option, you may select an existing layout to clone. If you create a page layout without cloning, your page layout will not include the standard sections whose

Existing Page Layout: Enrollment Layout

Page Layout Name: New Enrollment Layout

Save Cancel

3. Again, click New to create Re-Enrollment Layout:

- Clone existing layout → name it Re-Enrollment Layout → Save.
- Include fields like previous Enrollment reference (if any), Re-Enrollment Reason, Status, Fees, etc.
- Click Save.

Setup > Home > Object Manager

Enrollment

- Details
- Fields & Relationships
- Page Layouts**
- Lightning Record Pages
- Buttons, Links, and Actions
- Compact Layouts

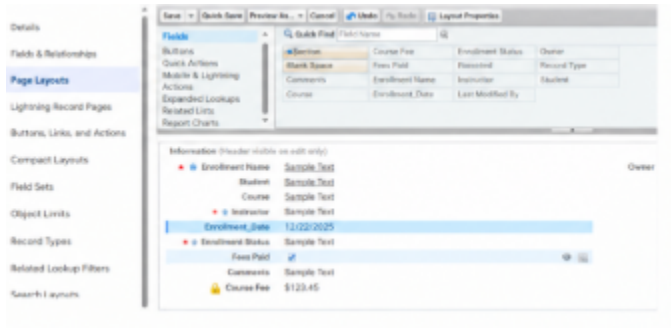
Create New Page Layout

As an option, you may select an existing layout to clone. If you create a page layout without cloning, your page layout will not include the standard sections whose

Existing Page Layout: Enrollment Layout

Page Layout Name: Re-Enrollment Layout

Save Cancel



4. Click Page Layout Assignment → Edit Assignment.
5. For each profile and record type:

- New Enrollment → use New Enrollment Layout.

6. Re-Enrollment → use Re-Enrollment Layout.

Click Save.

Activity 3: Enrollment Related Lists to Parent Objects:

Purpose:

To see all Enrollments from Student, Course, and Instructor records.

a) Student → Enrollment Related List

Go to Setup → Object Manager → Student__c.

Click Page Layouts → open the main layout.

In the palette, select Related Lists.

Drag Enrollments related list onto the layout.

Customize columns if needed → Save.

b) Course → Enrollment Related List

Go to Object Manager → Course__c → Page Layouts.

Open layout → add Enrollments related list.

Save.

c) Instructor → Enrollment Related List

Go to Object Manager → Instructor__c → Page Layouts.
Open layout → add Enrollments related list.
Save.

- **Milestone 10 – Automation with Flow**

Activity 1: Auto-Populate Enrollment Date (Record-Triggered Flow)

Purpose:

Set Enrollment__Date__c to today when Enrollment is created.

Steps:

1. Go to Setup → Flows → New Flow.
2. Select Record-Triggered Flow → Create.
3. Object: Enrollment__c.
4. Trigger: A record is created.
5. Optimize for Actions and Related Records → Done.

Select Object

Select the object whose records trigger the flow when they're created, updated, or deleted.

* Object

Enrollment

Configure Trigger

Trigger the Flow When:

- ☒ A record is created
☐ A record is updated
☐ A record is created or updated
☐ A record is deleted

Set Entry Conditions

Specify entry conditions to reduce the number of records that trigger the flow and the number of records that trigger the flow to conserve your org's resources.

Condition Requirements

None

6. Add Update Records element:

- Label: Set Enrollment Date.

- Use record that triggered the flow.
 - Field: Enrollment_Date__c → Value: \$Flow.CurrentDate.
7. Save as SetEnrollmentDate → Activate.

1. Activity 2: Send Email When Enrollment Approved

Purpose:

When Enrollment_Status__c = Approved, send email, alert admin, and set Student to Active.

Steps (high level for document):

1. Create Email Template and Email Alert for student notification.
2. Go to Flows → New → Record-Triggered Flow.
3. Object: Enrollment__c, Trigger on update.
4. Condition: Enrollment_Status__c Equals "Approved".

 Configure Start

Enrollment

Configure Trigger

Trigger the Flow When:

☐ A record is created
☐ A record is updated
☒ A record is created or updated
☐ A record is deleted

Set Entry Conditions

Specify entry conditions to reduce the number of records that trigger the flow and the number of times the flow is executed. Minimizing unnecessary flow executions helps to conserve your org's resources.

If you create a flow that's triggered when a record is updated, we recommend first defining entry conditions. Then select the **Only when a record is updated to meet the condition requirements** option for When to Run the Flow for Updated Records.

Condition Requirements

All Conditions Are Met (AND) ▼

Field	Operator	Value	
Enrollment Status X	Equals ▼	Approved X	

5. Add Get Records to fetch related Student__c.

 Get Records ×

* Label

Get Student

* API Name ⓘ

Get_Student

Description

Get Records of This Object

* Data Source

☐ Data Cloud Object

☒ Salesforce Object

* Object

Student

Filter Student Records

Condition Requirements

All Conditions Are Met (AND) ▼

Field

Operator

Value

Status X

Equals ▼

Aa Active X

+

 Add Condition

6. Add Action (Email Alert) to send email to student.

Enrollment Approved EmailAlert



* Label

Send Student Email

* API Name 

Send_Student_Email

Description

Use values from earlier in the flow to set the inputs for the "Enrollment Approved EmailAlert" email alert. To use its outputs later in the flow, store them in variables.

Set Input Values

 * Record ID 

 Triggering Enrolement__c > Record ID 

7. Add Action to notify Admin if needed.

8. Add Update Records to set Student__c.Status__c = "Active".


Update Records
×

* Label

Set Student Status Active

* API Name ⓘ

Set_Student_Status_Active

Description

* How to Find Records to Update and Set Their Values

☒ Use the enrollment record that triggered the flow
 ☐ Update records related to the enrollment record that triggered the flow
 ☐ Use the IDs and all field values from a record or record collection
 ☐ Specify conditions to identify records, and set fields individually

Set Filter Conditions

Condition Requirements to Update Record

None—Always Update Record ▼

Set Field Values for the Enrollment Record

Field

A Comments ×

←

Value

True ×

🗑️

9. Save as Enrollment Approved Action → Activate.

Activity 3: Auto-Assign Instructor Based on Course Category

Purpose:

Assign Instructor automatically based on Course category.

Rules:

Technical → Instructor A

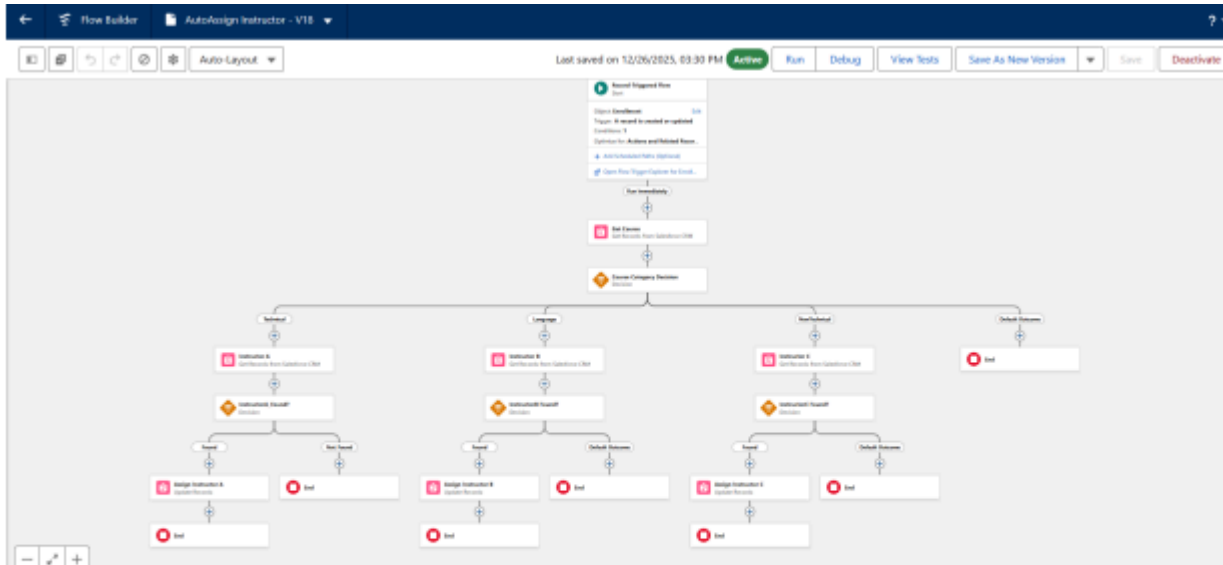
Language → Instructor B

Non-Technical → Instructor C

Steps:

1. Prepare Instructor__c records with codes (e.g., INSTR_A, INSTR_B, INSTR_C).
2. Create Record-Triggered Flow on Enrollment__c (created/updated).

3. Add Get Records to fetch related Course__c.
4. Add Decision element on Course__r.Category__c:
5. Technical / Language / Non-Technical outcomes.
6. For each outcome:
7. Get Instructor record (A/B/C) based on code or name.
8. Update Records: set Instructor__c = Get_Instructor_X.Id.
9. Save as AutoAssign_Instructor → Activate.



Activity 4: Auto-Create Task for Follow-Up

Purpose:

When Enrollment is “Requested”, create a follow-up task for Admin.
Steps:

1. Create Record-Triggered Flow on Enrollment__c.
2. Trigger: Record created or updated; condition Enrollment_Status__c = "Requested".

The screenshot shows the 'Configure Start' dialog in Salesforce. At the top, there's a green play button icon and the text 'Configure Start'. Below this, the 'Object' field is set to 'Enrollment'. The 'Configure Trigger' section has four radio buttons: 'A record is created', 'A record is updated', 'A record is created or updated' (which is selected), and 'A record is deleted'. The 'Set Entry Conditions' section includes a description and a note about condition requirements. Below this, the 'Condition Requirements' dropdown is set to 'All Conditions Are Met (AND)'. At the bottom, there's a table with three columns: 'Field', 'Operator', and 'Value'. The 'Field' column contains 'Enrollment Status', the 'Operator' column contains 'Equals', and the 'Value' column contains 'Requested'. There are also close buttons (X) for each field and value, and a trash icon for the entire row.

Field	Operator	Value
Enrollment Status	Equals	Requested

3. Add Get Records to fetch Admin user (by Username / Profile).
4. Add

Formula (Date): TODAY() + 2 for due date.

The screenshot shows the 'Get Records' configuration window in Salesforce. At the top, there is a red icon and the title 'Get Records'. Below this, there is a text input field containing 'Get Admin User' and a corresponding label 'Get_Admin_User'. A 'Description' field is also present. The section 'Get Records of This Object' includes a 'Data Source' section with two radio buttons: 'Data Cloud Object' and 'Salesforce Object', with 'Salesforce Object' selected. Below this is an 'Object' field containing 'User'. The 'Filter User Records' section has a 'Condition Requirements' dropdown set to 'All Conditions Are Met (AND)'. Below this is a table with three columns: 'Field', 'Operator', and 'Value'. The first row contains 'Username' (with a search icon), 'Equals', and 'geespam198430@agentforce.com' (with a search icon). At the bottom left of the table is a '+ Add Condition' button.

Get Records

Get Admin User Get_Admin_User

Description

Get Records of This Object

* Data Source

☐ Data Cloud Object

☒ Salesforce Object

* Object

User

Filter User Records

Condition Requirements

All Conditions Are Met (AND)

Field	Operator	Value
Username	Equals	geespam198430@agentforce.com

+ Add Condition

5. Add Create Records for Task:
6. Subject: Follow up on enrollment request.
7. OwnerId: Admin user Id.
8. WhatId: Enrollment Id.
9. ActivityDate: formula Today + 2.



Create a Record of This Object

* Object

Task

Set Field Values for the Task

Field

Due Date Only ×

Value

DueDateTodayPlus2 ×



Field

Description ×

Value

Auto-created follow-up for enrollment request



Field

Priority ×

Value

Normal ×



Field

Status ×

Value

Not Started ×



Field

Subject ×

Value

Follow up on enrollment request



Field

Related To ID ×

Value

Triggering Enrolement_c > Record ID ×

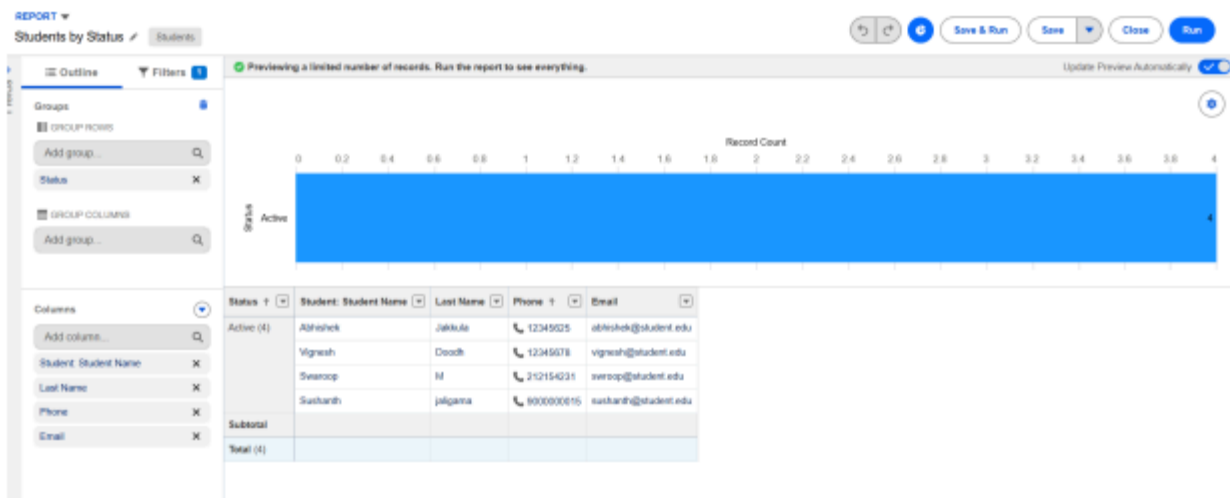


10. Save as Create_Followup_Task → Activate.

Milestone 11 – Reports:

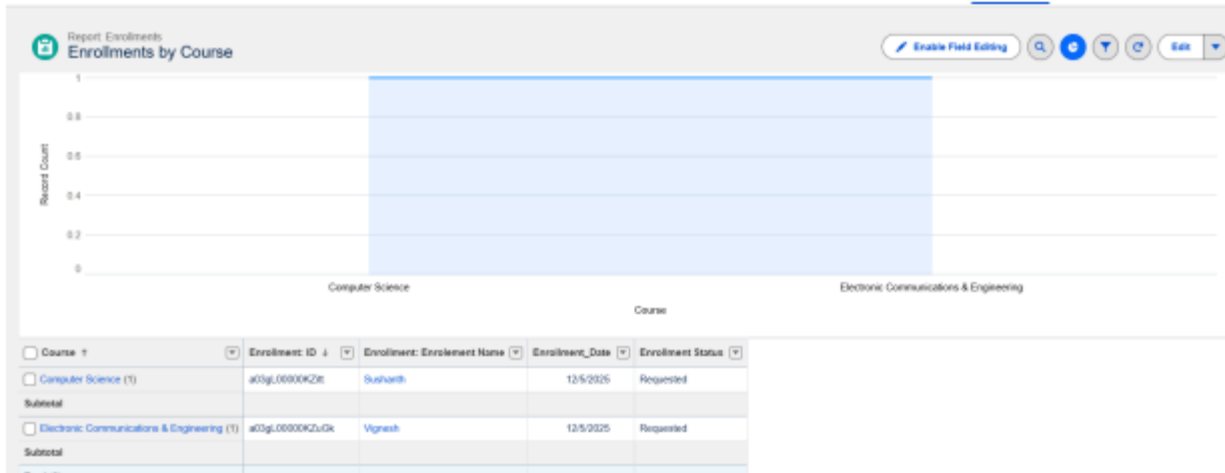
Activity 1: Create “Students by Status” Report

1. Go to **Reports** → **New Report**.
2. Report Type: **Students**.
3. Group rows by **Status__c**.
4. Add columns as needed (Name, Email, Phone).
5. Save as **Students by Status**.



Activity 2: Create “Enrollments by Course” Report

1. New Report → Report Type: **Enrollments**.
2. Grouprows by **Course__r.Name**.
3. Show Record Count per course.
4. Save as **Enrollments by Course**

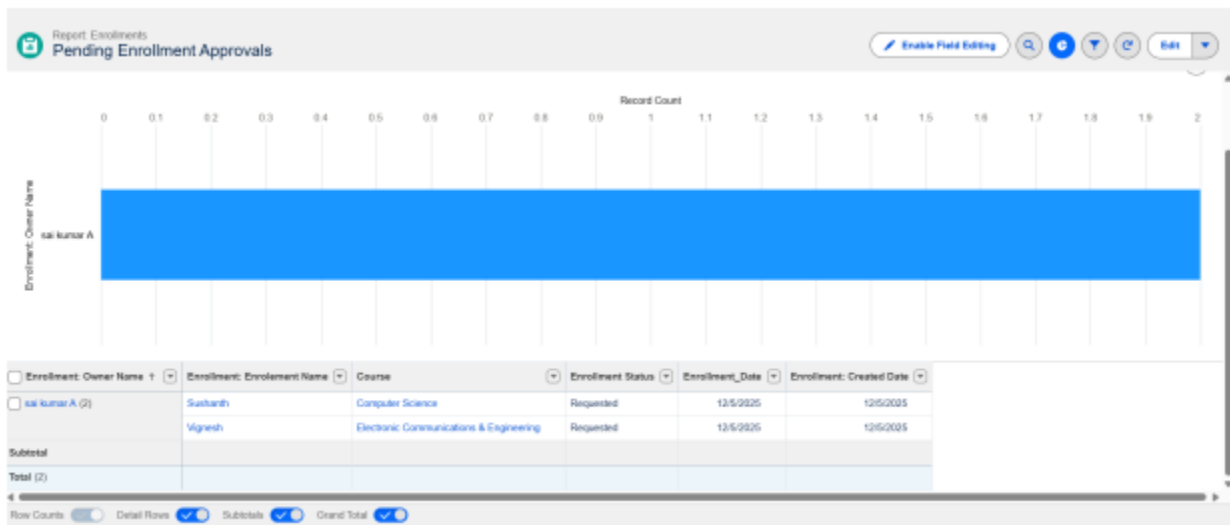


Activity 3: Create “Pending Enrollment Approvals” Report

1. New Report → **Enrollments** report type.
2. Filter: **Enrollment_Status__c = "Requested"**.
3. Add columns (Student, Course, Owner, Created Date).
4. Save as **Pending Enrollment Approvals**.

Activity 4: Create “Revenue Report (Fees Paid)” Report

1. New Report → **Enrollments** report type.
2. Filter: **Fees_Paid__c = TRUE**.
3. Group by **Course__r.Name**.
4. Summarize **Course_Fee__c** or similar fee field.
5. Save as **Revenue Report – Fees Paid**.

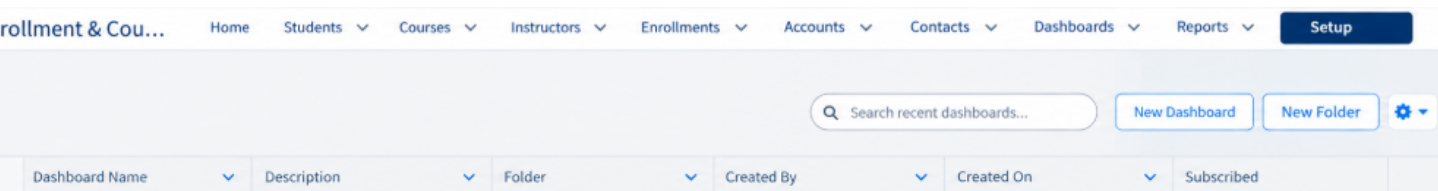


Milestone 12 – Dashboards:

Activity 1: Create “Student Management Dashboard”

Steps:

1. Go to **Dashboards** → **New Dashboard**.
2. Name: **Student Management Dashboard**.



3. Select folder → **Create**.
4. Select new Dashboard folder and Save.

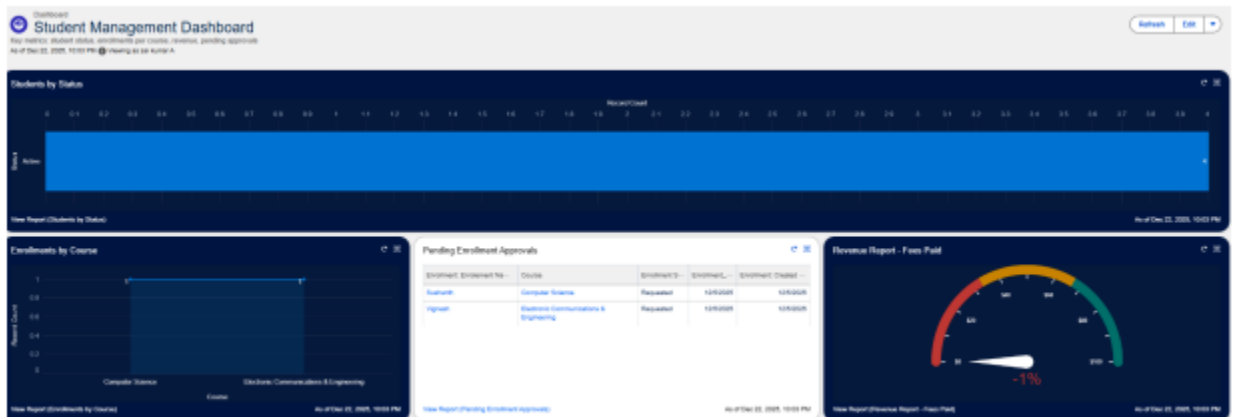
Add components:

- **Donut Chart** → using **Students by Status** report.
- **Bar Chart** → using **Enrollments by Course** report.
- **Gauge** → using **Revenue Report – Fees Paid** (sum of

revenue field).

- **Table** → using Pending Enrollment Approvals report.

4. Arrange components on canvas.



Click **Save** → **Done**.

Milestone 13 – Security Setup

Activity 1: Configure Profiles

):

1. Go to **Setup** → **Profiles** → **New** (or Clone Profiles Required):

- **Admin** – Full access.

- **Enrollment Officer** – Create/Update Enrollments.

- **Instructor** – Read-only access to assigned Enrollments.

2.Steps (example for **Enrollment Officer** fName:
Enrollment Officer).

3.In **Object Settings**:

- **Enrollment__c**: Read, Create, Edit.
- **Student__c** / **Course__c**: Read (and Edit if required).

4.Save.

Repeat similarly for **Instructor** profile with **Read-Only** on Enrollments

Activity 2: Create Permission Sets

1.Go to **Setup → Permission Sets → New**.

2.Create **Custom Object Access** (additional access to project objects if needed).

3.Create **Dashboard Editor** (Create/Customize Reports & Dashboards permissions).

4.Assign to relevant users.

Activity 3: Configure Sharing & OWD

Steps:

1.Go to **Setup → Sharing Settings**.

2. Set **Organization-Wide Defaults**:

- **Students (Student__c)**: Public Read/Write.
- **Courses (Course__c)**: Public Read/Write.
- **Enrollments (Enrollment__c)**: Private.

3. Save.

For Instructor access to assigned Enrollments, optionally:

- Add a **User lookup** on Enrollment (Instructor User).
- Use a simple **Record-Triggered Flow** to set **OwnerId = Instructor User** so they can see only their own records.

7. FUNCTIONAL & PERFORMANCE TESTING

During the testing phase, screenshots were captured from the Salesforce user interface to verify the proper functioning of all implemented features of the Student Enrollment & Course Management System (SECMS). These include student, course, instructor, and enrollment record creation; validation rule checks for mandatory fields, eligibility, and enrollment constraints; verification of record type behavior for different course categories; flow execution for automated enrollment approvals and seat availability checks; and Apex trigger functionality to ensure automatic confirmation and record updates. Additionally, screenshots of generated reports and dashboards were documented to validate accurate data visualization, enrollment tracking, and performance monitoring. Submit the Screenshots while testing.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Centralized data
- Automated workflows
- Real-time reporting

Disadvantages

- Salesforce dependency
- Requires initial configuration effort

9. Conclusion:

The Salesforce Student Enrollment & Course Management System successfully streamlines the entire student academic lifecycle—from student registration and course creation to enrollment management, instructor assignment, approvals, revenue tracking, and reporting. By leveraging Salesforce’s automation tools such as validation rules, flows, approval processes, and dashboards, the system improves data accuracy, operational efficiency, and decision-making. This project demonstrates how Salesforce can transform manual academic processes into a scalable, reliable, and user-friendly digital solution.

10. FUTURE SCOPE

- Payment gateway integration
- Student self-service portal
- Analytics-driven insights
- Mobile access support

11. APPENDIX

- Source Code: Salesforce Configuration
- Dataset: Salesforce Objects
- GitHub / Demo: Salesforce Developer Org

Thank You